Some basic arguments about the representations that determine and instantiate stress
Claire & Jonah's stress n' structures seminar, week 1

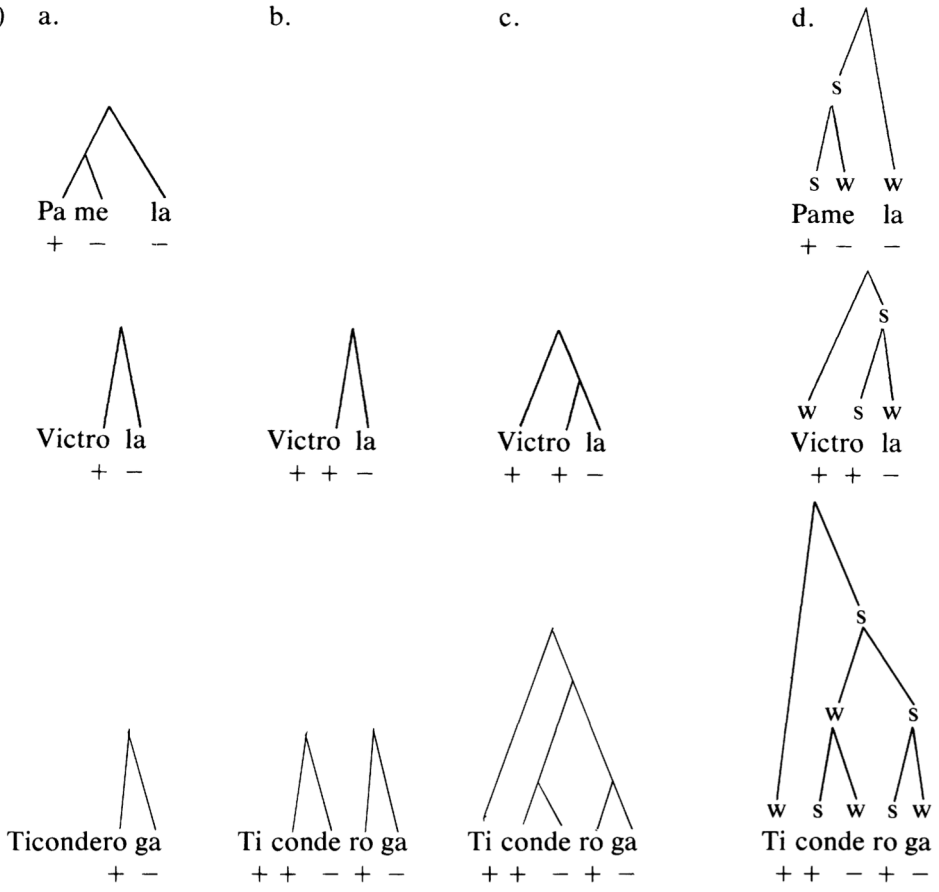
Selkirk 1980:

- Stress is a syllable that is dominated by a foot and is not labeled w
- Foot shapes are dictated by a limited number of templates
- These templates are part of URs, or are imposed on URs before any phonology happens
- The feature [stress] is eliminated from the theory

Starting from the theory one theory before the current theory (simplified Liberman & Prince)…

(5) *English Stress Rule* (a standard version)
[+syll] → [+stress] / [. . . ___ C_0(V(C)) (V C_0)]

(6)



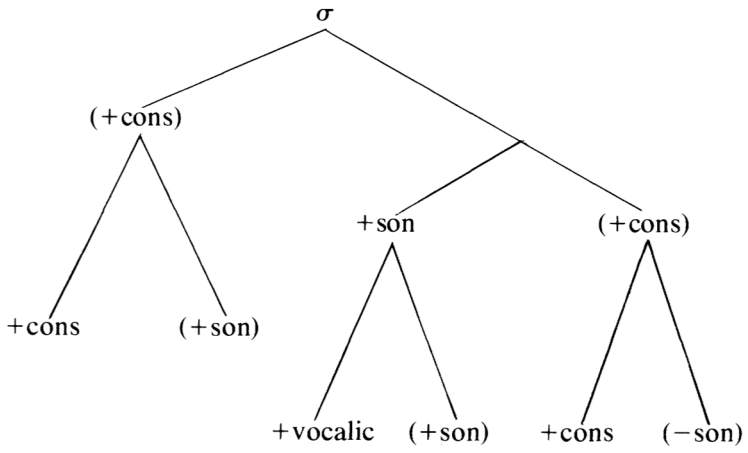
(7) *LCPR*
Given a pair of nodes [N_1 N_2], N_2 is *s* iff it branches.

So this theory uses a [stress] feature and tree structures, where the tree structures are mostly just there to encode the relative prominence amongst [+stress] syllables.
Selkirk's proposed alternative:

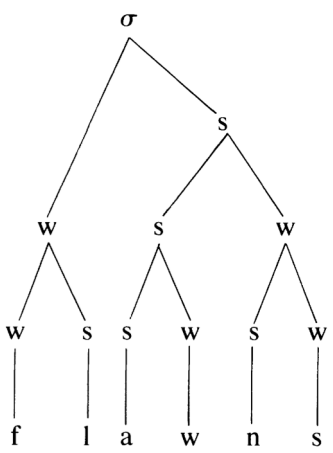
formed prosodic words. In general, it will be said that for any prosodic category K_i, be it above or below the level of prosodic word, a triplet $K_i = (C_{K_i}, P_{K_i}, D_{K_i})$ is to be defined, where C_{K_i} are the *principles of construction* specifying the nature of structure internal to K_i (constituency in terms of other categories, direction of branching, etc.), P_{K_i} are the *principles of prominence* specifying the *s/w* relations of the subtrees constituting K_i, and D_{K_i} is the *syntactic domain* within which these well-formedness conditions are met. In what follows, the hypothesis will be entertained that these well-formedness conditions are satisfied at underlying phonological representation, which is to say that prosodic structure is "established" or defined from the start in a phonological derivation (which, given our approach, is tantamount to saying that "stress" is the "first rule" to "apply" in the phonology).

Syllable-level principles of construction:

(9) *English Syllable Template*[6]

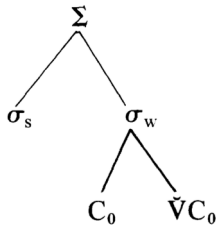


(8)



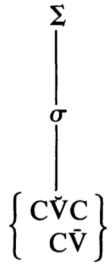
Foot-level:

(33) *The Bisyllabic Stress Foot*



(21) *The Monosyllabic Stress Foot*

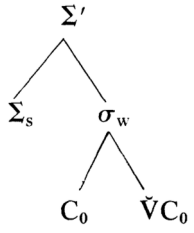
a.[11]



b. initial in D_Σ, Σ



(34) *The Stress Superfoot*



PWd-level:

The category prosodic word is defined as being constituted of a sequence of one or more Σ or Σ′ joined in a right-branching structure (= the principles of construction, C_{ω}). The Lexical Category Prominence Rule (7) specifies the *s/w* relations in the tree above the level of Σ or Σ′. That is, the LCPR represents the principles of prominence,

The syntactic domain within which each of these generalizations holds: the **stem**

Let's try to build structures for a bunch of attested and unattested word types…

[pásəm] 'possum' [tætú] 'tattoo' [ælə͡ɹ] 'ally' [kəzú] 'kazoo' (!)

[kænədə] 'Canada' [bənænə] 'banana' [ʃɛvɹəle͡ɪ] 'Chevrolet' *[ʃəvəlje͡ɪ] 'chevalier'

[əlɛktɹək] 'electric' [gæləksi] 'galaxy' [kærəməɹæn] 'catamaran' *[kærəməɹən]

(We should probably also try *[bətætú] or *[kæbətætú])

Main argument against the Latin-stress-rule/SPE-based alternative in (5): **exceptionality**. Here, *gálaxy* and *banána* are just normal words, whose lexical specifications happen to differ from *eléctric* and *Cánada*. In the rule-based alternative, they'll have to be marked with an exception feature (even weirder, it would have to be an exception to *certain expansions* of the parentheses notation).

Some of the additional rules Selkirk needs…

Not specified in any detail, but vowels in weak syllables (*not* weak feet) reduce to schwa.

This will then be fed by various other rules that defoot certain syllables. E.g. for unstressed initials:

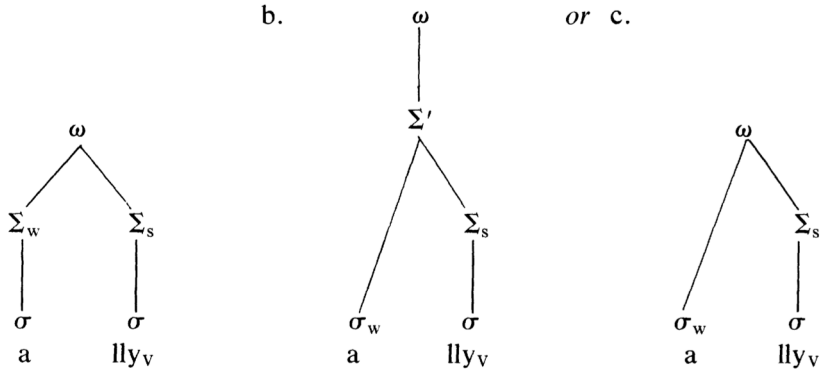
(27) *Defooting*

$\omega(\Sigma_w(\sigma(C\check{V})_\sigma)_{\Sigma_w} \ldots)_\omega \Rightarrow \omega(\sigma_w(CV)_{\sigma_w} \ldots)_\omega$

So all of these start out life as footed syllables.

Input Output b or c (she remains agnostic on 'trapped lights')

(28) a. b. *or* c.



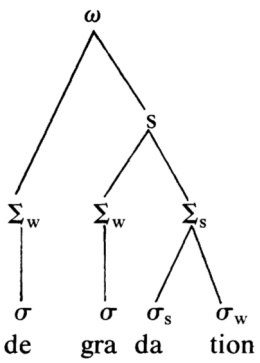
Also, we'll need to do medial reduction of certain tense-voweled syllables:

(41) *Medial Defooting*[17]

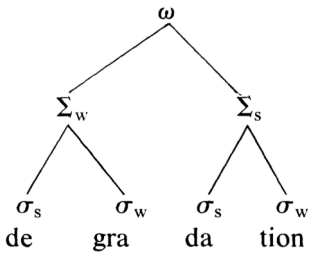
$\omega(P\ \Sigma_w(\sigma(CV)_\sigma)_{\Sigma_w}\ Q)_\omega \Rightarrow \omega(P\ \sigma_w(CV)_{\sigma_w}\ Q)_\omega$

Condition: P, Q ≠ ϕ

(39)



(42)



Finally, we'll also need some *tensing* rules, for word-final and prevocalic schwa/lax vowels:

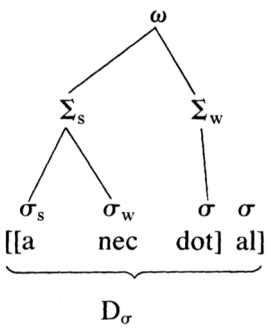
(16) $\begin{bmatrix} +\text{vocalic} \\ -\text{low} \end{bmatrix} \rightarrow [+\text{tense}] \,/\, {}_{\omega}(\ldots \underline{\quad})_{\omega}$

Implicitly, also /_V within the PWd domain. So at the time they're footed, these vowels would not be [+tense].

She doesn't go into too much detail about cyclicity in this paper, and I don't really want to either at this point in the course. But here are a couple ideas:

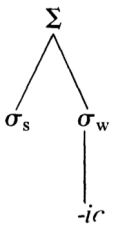
Intermediate stage

(49)



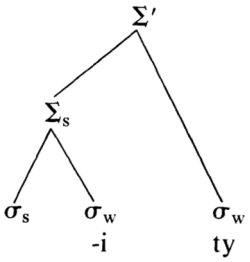
UR

(50)

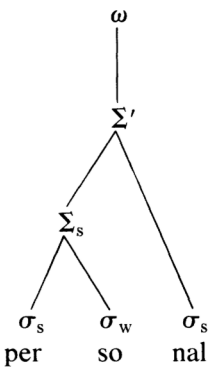


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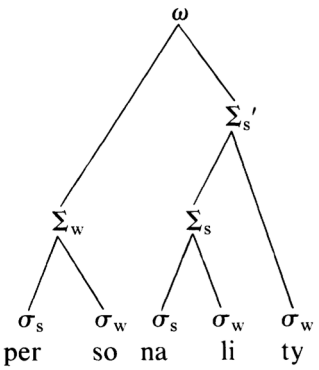
(52)



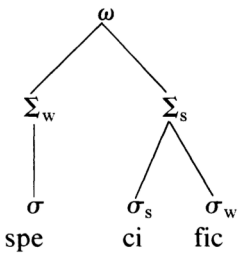
(53) a.



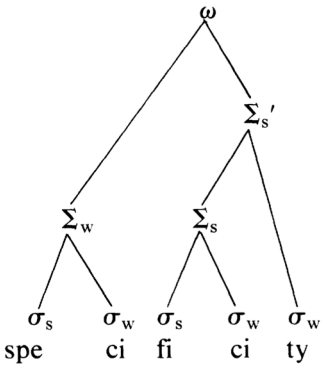
b.



(54) a.



b.



Shorter discussions of the other two papers

McCarthy 1982: Infixation in English (not original to him, *contra* what I said in the email):

(1) Mononga-fuckin-hela
Ala-fuckin-bama
fan-fuckin-tastic (M)

(4) a. fan-fuckin-tastic *fant-fuckin-astic *fa-fuckin-ntastic
b. Du-fuckin-brovnik *Dubr-fuckin-ovnik ?Dub-fuckin-rovnik
c. in-fuckin-stantiate *inst-fuckin-antiate ?ins-fuckin-tantiate *i-fuckin-nstantiate

(5) amalga-bloody-mated (M)
emanci-motherfuckin-pator (M)
every-bloody-body (M)
handi-bloody-cap (M)
hypo-bloody-crite (M)
kinder-goddamn-garten (M)
Lauder-damn-dale (M)

(8) a. Kentucky → Ken-fuckin-tucky (0 1)
b. Nebraska → Ne-fuckin-braska (0 1)
c. Hawaii → Ha-fuckin-waii (0 1)

Trying to do this without feet:

(9) Expletive Infixation (Syllabic version)
[X Q V Y]
[+stress]
1 2 3 4 → 1 EXPLETIVE 2 3 4
Condition: Q does not contain syllable boundary.

(10) Expletive Infixation (Metrical version)
X [Y]$_\Sigma$
1 2 → 1 EXPLETIVE 2

With feet:

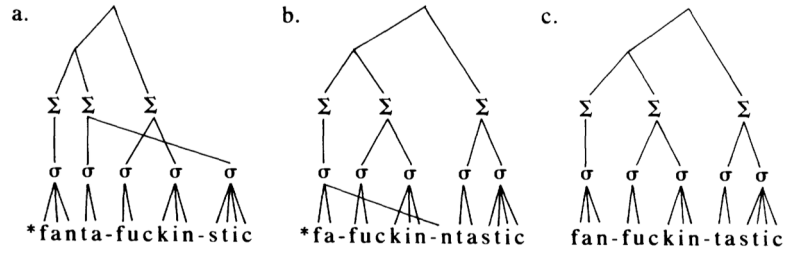


FIGURE 3.

Key POS argument:

found in McCarthy 1981.) For sociological reasons, many speakers of English are not exposed to primary data with infixed expletives until adolescence. Even then, the data are quite degenerate, consisting in most cases of just a few types like *fan-fuckin-tastic*. Despite this, grammaticality judgments are quite sharp, with a remarkable degree of reproducibility—a point also made by McMillan. It is difficult to reconcile these two facts if we imagine Expletive Infixation to be just another morphological rule, presenting some constellation of arbitrary phonological conditions to the language learner.

An extension to the description that segmental rules can't handle, but recursive feet can:

(11) a. $\overset{3}{\text{Tatamagouchee}}\ \overset{1}{}$ →
Tata-fuckin-magouchee (or Tatama-fuckin-gouchee)

b. $\overset{3}{\text{Winnipesaukee}}\ \overset{1}{}$ →
Winni-fuckin-pesaukee (or Winnipe-fuckin-saukee)

c. $\overset{3}{\text{Kalamazoo}}\ \overset{1}{}$ →
Kala-fuckin-mazoo (or Kalama-fuckin-zoo)

Claim: the domain of tapping converges on the same type of structure:

(12) repetitive
a. repe[t^h]i[t^h]ive
b. repe[D]i[t^h]ive
c. repe[D]i[D]ive

Selkirk-style feet allow us to explain both of these generalizations:

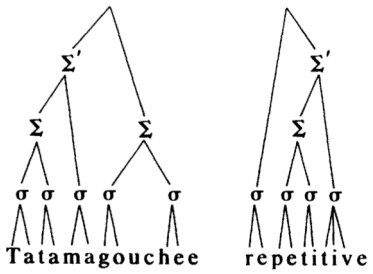


FIGURE 5.

One more thing:

(15) un-fuckin-believable (M)
un-fuckin-derivable
un-fuckin-collectable
un-fuckin-dissuaded
*im-fuckin-possible
*ir-fuckin-responsible
…

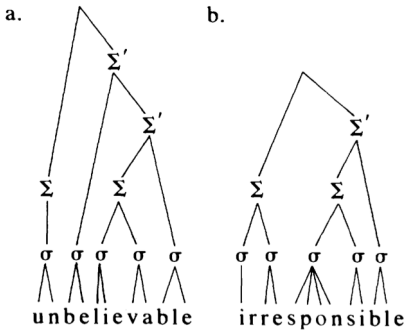


FIGURE 8.

Hayes 1982: Let's keep the Selkirk-style foot theory but also allow for rules that render entire classes of syllable **invisible to footing** rules (*extrametrical*).

Classical Arabic stress:

(1979a,b), the stress pattern of Classical Arabic (and some of its modern descendents) is based on the distinction of light (CV), heavy (CVV and CVC), and superheavy (CVVC and CVCC) syllables. Stress falls (a) on superheavy syllables, which may occur only in phrase-final position; (b) otherwise, on the rightmost nonfinal heavy syllable; (c) otherwise, on the initial syllable. Some examples are as follows:

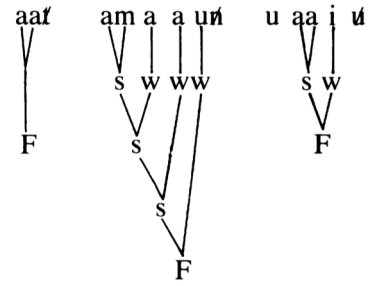
(6) kaatibáat 'writer (fem. pl.)'
yušáariku 'he participates'
mámlakatun 'kingdom (nom. sg.)'
kátaba 'he wrote'

Claim: final syllables are demoted SH → H, H → L because the final segment is invisible to footing.

(7) *Final Segment Extrametricality*
$[+\text{seg}] \rightarrow [+\text{ex}] \; / \; ____ \;]_{\text{word}}$

(8)

kaatibaat	mamlakatun	yušaariku	
aa i aat	am a a un	u aa i u	rhyme projection
aa i aat	am a a un	u aa i u	Final Segment Extrametricality
aa i aat	am a a un	u aa i u	Foot Construction



Similar example in Hopi, but with one initial iamb and final syllable extrametricality.

Latin stress rule (also maybe English, a different version of classical Arabic):

(20) a. X (x)
s w

b.

	e ec us	e ee it	e i it	Extrametricality	reféctus	refec it	réficit	other rules
	e ec us	e ee it	e i it	Foot Construction	ws w	ws w	s ww	
			s w		s	s	s	

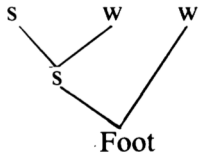
Mopping up the 'trapped' syllables:

(21) *Stray Syllable Adjunction* (SSA)
Adjoin a stray syllable as a weak member of an adjacent foot.

Finally, we come to English stress. Worth checking out this entire passage:

The productivity of certain aspects of English stress is unquestionable. LP point out, for example, that forms like *pódectal* and *pónitode* ([pánəDôwd]), which violate the rules of foot construction, are not only missing from the corpus, but also sound strange to native speakers. Notice also that the Russian words *Nínotchka, bábushka* were borrowed into English with penultimate stress: the original Russian stressings would be very strange for English words. However, these cases do not necessarily show that there is a set of rules deriving stress in English. Selkirk (1980) has suggested that speakers' knowledge about English stress is embodied only in their knowing the maximum possible size of a metrical foot, the template being roughly as shown in (22):[3]

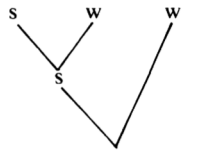
(22) C_0VC_0 $(C_0\check{V})$ $(C_0\check{V}C_0)$



Pódectal, póni[*D*]*ode, Nínotchka*, and *bábushka* all constitute metrical feet that could not be encompassed by the template (22). They thus can be ruled out without a pho-

[3] Selkirk's template is actually of the form (i),

(i) C_0VC_0 $(C_0\check{V}C_0)(C_0\check{V}C_0)$



which incorrectly predicts *pódectal, Nínotchka* to be normal English words. I assume that this is simply an oversight.

I don't think this was an oversight, I think they disagree about the productivity of the generalization here. In Selkirk's framework, CVC is a licit w syllable in a foot, and a 3-syllable 'superfoot' can include CVC as either the second or third syllable. I don't know about the status of *pódectal* (seems weird to me) or *nínochka* (seems fine, but I speak some Russian). But *gálaxy*, *cílinder, héraldry*, etc. all exist. Parallel to this disagreement, Hayes notes that they also disagree on the productivity of antepenultimate vs. penultimate stress in >2-syll words.

This is the crux of the disagreement, and I don't think it's ever been resolved:

Several considerations argue against Selkirk's claim, however: (a) Historical change: we will see below that the various changes that have occurred in English stress since the eighteenth century can be explained as the removal of irregular diacritic markings from lexical representations. The explanation crucially requires that we invoke a phonological derivation, in that the relevant words were well-formed in Selkirk's sense before the change as well as after. (b) Experiment: Trammell (1978) has shown that when subjects are asked to read unfamiliar words of English, for the most part they prefer pronunciations having feet that are maximal according to the rules that follow. (c) The cycle: in Hayes (1981, 163–164), it is argued that there are serious reasons for doubting Selkirk's explanation of the phonological cycle as being merely the result of the language's history. Only a derivational account of English stress appears to be adequate to explain cyclic phenomena.

Anyhow, Verbs and adjectives:

(23)

obéy	torménт	astónish
atóne	usúrp	devélop
divíne	robúst	cómmon
discréet	ovért	illícit

(24) *Consonant Extrametricality*

$[+\text{cons}] \rightarrow [+\text{ex}] \,/\, \underline{\quad} \,]_{\text{word}}$

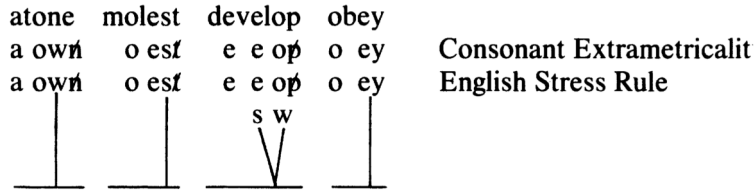
(25) *English Stress Rule*

At the right edge of a word, form a maximally binary foot on the rhyme projection using the template X (x)—that is, the right node of a branching foot



must dominate a nonbranching rhyme. Label feet *s w*.

(26)



Nouns are Latin:

In the nonfinal syllables of nouns having stressless final syllables, the situation is considerably more regular: stress is generally assigned to a heavy penult, while the antepenult receives stress if the penult is light:

(32)

América	Arizóna	agénda
díscipline	factótum	appéndix
lábyrinth	elítist	amálgam

(33) *Noun Extrametricality*

$\text{Rhyme} \rightarrow [+\text{ex}] \,/\, \underline{\quad} \,]_{\text{N}}$

the stressing of *labyrinth, Arizona*, and *agenda*:

(34)

a i inθ	a i ow a	a en a	Consonant Extrametricality
a i ɨnθ	a i ow ə	a en ə	Noun Extrametricality
a i ɨnθ	a i ow ə	a en ə	English Stress Rule
s w			
a i ɨnθ	a i ow ə	a en ə	Stray Syllable Adjunction
s w w	s w	s w	
s			
lábyrinth	Àrizóna	agénda	other rules
s w w	s ws w	w s w	
s	w s	s	

There is **a lot** more in this paper about various cleanup rules, stress retraction, the cycle, exceptions, lexical strata, etc. I'm not going to go through all of it. Main point here is about the tension between more lexical theories where more facts about stress are just arbitrary choices from a menu of options, and more phonological theories where more more facts are systematic but there are a ton cleanup rules, cyclic effects, and/or exception diacritics.

One more thing: anytime somebody uses Monòngahéla to illustrate some systematicity of English stress, remember that the river passes through Mònongália County, WV on its way North to Pittsburgh. As far as anybody can determine, these are loans from the same Lenape word at two different times.